

Composition Pair B1.02 + B1.17 — The Co-Required Pair Where the Three-Level Structure Requires Purpose-Defined Role Determination to Assign Entities to Levels, and Relational Role Determination Requires a Structured Level System to Assign Entities Into, Making Each Commitment Architecturally Inoperable Without the Other

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026).

Abstract

This note is the fourth in Phase B4 of the CKS derivation note series and the third composition pair note. It formalizes the co-required relationship between B1.02 (three architectural levels: cell, aspect, Self) and B1.17 (relational roles: level membership determined by functional purpose rather than intrinsic technical type). The two commitments are co-required rather than merely complementary: B1.02 provides the structured assignment space — three levels, each with distinct governance scope — that B1.17's determination mechanism needs in order to have somewhere to assign entities; B1.17 provides the population mechanism — purpose-defined determination of which level an entity belongs to — that B1.02's levels need in order to be populated by appropriately-governed entities rather than by default technical classification. Without B1.02, B1.17's determination produces a result with no architectural destination. Without B1.17, B1.02's levels exist but are populated by technical type rather than by functional purpose, defeating the governance motivation for distinguishing the levels in the first place. The operational intersection of the two commitments is B2.85 purpose-defined level determination, where B1.17's criterion meets B1.02's structure in practice; the dynamic continuation is B2.87 evolution-triggered role change, where function changes during an entity's lifecycle trigger level reassessment within B1.02's structure under B1.17's criterion. Absent B1.02 alone, the system collapses into the Flat Architecture anti-pattern (B3.03). Absent B1.17 alone while B1.02 is present, the system instantiates the Intrinsic Type Assignment anti-pattern (B3.18), in which structurally correct levels are populated by entities whose governance scope mismatches their actual function.

1. Pair identification and series position

Pair: B1.02 (three architectural levels) + B1.17 (relational roles — level membership by functional purpose).

Relationship type: Co-required pair. Neither commitment is operationally effective without the other. The relationship is not one-directional enabling, where one commitment unlocks the other, but mutual — each commitment creates a gap that only the other can fill.

Series position: B4.04 is the fourth Phase B4 note and the third composition pair note. The two preceding pair notes are B4.02 (B1.01 + B1.06: instinct/reasoning separation and governed origination) and B4.03 (B1.01 + B1.07: instinct/reasoning separation and mating as governed content combination). The note following this one, B4.05, will address B1.02 + B1.20 (three-level structure and recursive governance). The present note stands between those two: it formalizes how the three-level structure acquires its entities before B4.05 addresses how governance propagates recursively across the populated levels.

2. The two commitments individually

B1.02 — Three architectural levels. The source paper's Claim 2 (§5.2) establishes three structurally distinct levels: the cell, the atomic unit Paper 1 defends; the aspect, a coordination arrangement of cells serving a particular purpose; and the Self, the integrated whole holding multiple aspects under unified governance. The three levels are not a convenience taxonomy. Each carries a distinct governance scope: cell-level entities are governed by Paper 1's six commitments per B2.99; aspect-level entities are governed under the aspect scope per B2.100; Self-level entities are governed at the scope that unifies all aspects per B2.101. The structural distinctness of the levels is what makes distinct governance scopes meaningful — if cell and aspect and Self are not architecturally separate, their governance commitments blur together and the per-level work the source paper does in §5.2 and §5.5 is undone.

B1.17 — Relational roles. The source paper commits (§5.2) that level membership is determined by functional purpose rather than by intrinsic technical type. The same underlying CKS artifact can participate as a cell in one arrangement and as a component of an aspect in another, depending on what the structure is for. An artifact's level assignment is not fixed by how it is implemented — whether it is a substrate, an orchestration rule set, a grouped collection of substrates, or any other technical object — but by the governance function it performs in a given deployment arrangement. This commitment is what the source paper calls relational structural roles: the role-instances inhere in the bearer per the structural arrangement that calls on them, not in the bearer intrinsically.

These two commitments are each independently statable but neither is independently operable.

3. Enabling direction one: B1.02 requires B1.17

The three-level structure establishes cell, aspect, and Self as distinct architectural levels. But establishing three levels does not tell a designer which deployment entities belong at which level. For any actual deployment, the levels must be populated — specific entities must be assigned to cell level, aspect level, or Self level.

Without B1.17, the only available population mechanism is technical type. An entity is assigned to cell level because it looks like a cell (it has substrates and orchestration rules), to aspect level because it looks like an aspect (it groups cells), to Self level because it looks like a Self (it holds aspects). Technical type assignment is not a formal fallback; it is simply what happens when no determination criterion is specified. Designers default to what they can see in the implementation.

Technical type assignment defeats the governance motivation for the three levels. The reason the source paper introduces distinct levels — rather than treating all entities as undifferentiated cells — is precisely that different governance functions require different governance scopes. An entity performing coordination across multiple cells serves a different governance function than an entity performing a specific informational task, and the architecture's ability to apply distinct governance to each depends on having distinguished them by function, not by appearance. If levels are populated by what entities look like technically rather than by what governance function they perform, the level distinctions are cosmetic: the architecture has three labeled boxes but no reliable mechanism for ensuring the right entities are in the right boxes.

B1.17 provides the needed determination criterion. By specifying that level membership follows functional purpose, B1.17 gives B1.02's levels a principled population mechanism. The three levels exist in order to support distinct governance scopes; B1.17 ensures that what populates each level is determined by whether the entity's function fits that scope.

4. Enabling direction two: B1.17 requires B1.02

The relational roles commitment specifies that level membership is determined by functional purpose. This determination process produces a level assignment for each entity: this entity performs a cell-level function; that entity performs an aspect-level function. The determination is only meaningful if there is a structured level system to assign into.

Without B1.02, B1.17's determination produces a result with no architectural destination. The criterion — "this entity performs coordination-across-cells functions" — remains well-defined, but the output of applying the criterion cannot be placed anywhere. There is no aspect level to assign the entity to, because B1.02 is the commitment that establishes aspect as a distinct level in the first place. The determination is analytically coherent but architecturally inert: it produces a label ("this entity performs aspect-level functions") that the architecture has no structure to honor.

More precisely, without B1.02, B1.17 collapses into an informal labeling practice. Designers may informally note that some entities perform coordination functions while others perform task functions,

but without B1.02's level structure, there is no architectural mechanism to apply distinct governance scopes to these informally labeled groups. The governance implications of the level assignment — which Paper 1 commitments apply, what lifecycle operations are available, which evolution mechanisms operate — are defined per level by B1.02's structure. B1.17 produces assignments to that structure; without the structure, the assignments are notations without operational force.

5. Mutual dependency: the assignment space and the population mechanism

The co-requirement follows from the structure of what each commitment provides.

B1.02 provides the assignment space. The three levels — cell, aspect, Self — are the structured space within which level determination operates. Level governance scopes (B2.99 at cell level, B2.100 at aspect level, B2.101 at Self level) are meaningful because the levels are structurally distinct per B1.02. The assignment space has content: each level has a defined governance scope, lifecycle operations appropriate to its scope, and evolution mechanisms that operate at its scope. An entity assigned to cell level by B1.17 inherits all of that structure. The assignment space is not a set of empty labels; it is an architecturally articulated structure into which B1.17's determination places entities.

B1.17 provides the population mechanism. B1.02's levels are empty architectural structures until B1.17 provides a criterion for placing entities into them. B2.85 — purpose-defined level determination — is the operational expression of B1.17's population work: it specifies that the determination criterion asks what governance function the entity performs, and that the answer to that question determines level placement. Without B1.17, the population mechanism defaults to technical type; with B1.17, the population mechanism is principled and aligned with the governance motivation for having distinct levels.

The operational intersection of the two commitments is B2.85. When a deployment designer evaluates a new entity and asks "what level does this entity occupy?", the answer is produced by applying B2.85's criterion (what governance function does this entity perform?) within B1.02's level structure (cell, aspect, or Self). The two commitments are both active in that single determination step, and neither produces the correct answer without the other.

6. Operational integration

The pair operates at two moments in an entity's existence.

At deployment design and birth (B1.09). Each entity is assessed per B2.85's determination criteria: what governance function does this entity perform? Does it handle a specific informational task (cell level)? Does it coordinate cells serving a particular purpose (aspect level)? Does it hold multiple aspects as facets of one unified whole (Self level)? The determination produces a level assignment within B1.02's structure. The assigned level then determines which governance scope applies — which

Paper 1 commitments are active, which lifecycle operations are available, which evolution mechanisms are appropriate. The pair is jointly active from an entity's origination.

Through lifecycle evolution (B1.14, B2.87). The architecture is not static. As entities mature through directed selection (B1.14), an entity's governance function may change substantially — a cell that originally handled a specific informational task may evolve into a coordination arrangement that performs aspect-level functions. B2.87 evolution-triggered role change is the operational expression of B1.17's criterion applied to this dynamic situation: when an entity's function changes significantly, B1.17's determination criterion is re-applied, and if the result no longer matches the entity's current level assignment within B1.02's structure, the assignment changes. The pair creates a dynamic architecture in which entities occupy levels that reflect their current function, not their function at origination.

This dynamic coupling matters because the governance implications of level assignment are not administrative labels — they are operationally active. An entity reassigned from cell level to aspect level inherits aspect-level governance scope, aspect-level lifecycle operations, and aspect-level evolution mechanisms. The pair's joint operation through B2.87 ensures that governance scope tracks function continuously, not only at birth.

7. Co-presence requirement

B1.02 without B1.17. When three-level structure is present but relational role determination is absent, entities are assigned to levels by technical type rather than functional purpose. The levels are structurally present but populated by default classification. An entity that looks technically like a cell is treated as a cell regardless of whether it actually performs a cell-level governance function; an entity that looks technically like an aspect is treated as an aspect regardless of whether its governance function is at cell scope. The result is the Intrinsic Type Assignment anti-pattern (B3.18): the level structure is architecturally correct, but the governance scopes applied to entities are misaligned with the entities' actual functions. The purposes for which the three levels were introduced — distinct governance for distinct functions — are not served.

B1.17 without B1.02. When relational role determination is present but three-level structure is absent, level determination produces results with no architectural destination. Designers may correctly identify that some entities perform coordination functions and others perform task functions, but there is no structured level system to assign them into. The architecture is effectively flat — all entities are treated as cells or as undifferentiated components — and B1.17's determinations are informal notations without governance force. This is the Flat Architecture anti-pattern (B3.03): the determination criterion exists but the structure it assigns into does not, so distinct governance scopes cannot be applied.

Full co-presence. B2.85 purpose-defined level determination (operational expression of B1.17) populates B1.02's levels with entities whose governance scope matches their functional purpose. Each

entity at each level has appropriate governance applied per the level structure. B2.87 evolution-triggered role change maintains this alignment through the entity's lifecycle. The pair jointly produces a dynamic architecture where level assignment tracks function and governance tracks assignment.

8. Detection

Four indicators support detection of pair completeness.

B2.09 level-distinguishability tests. Do the tests that distinguish entities at each level use functional criteria — criteria about what governance function the entity performs — rather than technical criteria about what the entity is implemented as? Functional criteria indicate B1.17 is active; technical criteria indicate it is absent.

B2.85 determination records. Are level assignments documented with purpose-defined determination rationale? A determination record specifies what governance function the entity performs and how that function maps to a level within B1.02's structure. Absent determination records, level assignments are likely based on technical type.

B2.88 relational roles verification. Are determination records present for each entity in the deployment, not only for a selected subset? Partial determination coverage indicates that B1.17 is applied inconsistently and that some entities are assigned by default rather than by purpose.

Pair completeness test. Do level assignments (B1.02) derive from purpose-defined determinations (B1.17)? A deployment satisfies this test if every entity's level assignment can be traced to a B2.85 determination record specifying the governance function that grounds the assignment. A deployment fails this test if any entity's level assignment is explained only by its technical implementation type.

9. Failure modes

Flat Architecture (B3.03) — when B1.02 is absent. Without three-level structure, all entities occupy a single undifferentiated level. B1.17's determination may still be applied — designers may identify which entities perform coordination functions versus task functions — but the determination produces no architectural consequence. Governance scope cannot vary by level if there are no levels. The system does not inherit distinct governance for distinct functions; all entities are governed uniformly at whatever scope the flat architecture specifies.

Intrinsic Type Assignment (B3.18) — when B1.17 is absent while B1.02 is present. The three levels exist and are populated, but the population criterion is technical type rather than functional purpose. Entities are assigned to cell level because they look like cells, to aspect level because they look like aspects, to Self level because they look like Selves. The governance scopes the levels carry are applied

to entities based on their implementation appearance rather than their governance function. This failure mode is subtler than Flat Architecture: the architecture looks correct structurally, but the governance alignment that motivates the three-level structure is absent. Entities performing coordination functions may be governed at task scope, and entities performing task functions may be governed at coordination scope, because their technical appearance determined their assignment rather than their function.

10. Conclusion

The three-level structure (B1.02) and relational role determination (B1.17) are co-required commitments. Each provides what the other needs to be operationally effective. B1.02 provides the structured assignment space — three levels with distinct governance scopes — that B1.17's determination needs in order to produce level assignments with architectural force. B1.17 provides the population mechanism — purpose-defined determination — that B1.02's levels need in order to be populated by entities whose governance scope matches their function.

The operational expression of the pair is B2.85, where purpose-defined determination places entities into the three-level structure at deployment birth. The dynamic continuation is B2.87, where evolution-triggered role change applies the same determination criterion through an entity's lifecycle, ensuring that governance scope tracks function even as the entity evolves. The pair jointly produces an architecture in which level assignments are principled, governance scopes are functionally aligned, and both properties are maintained dynamically through directed selection.

Absent the pair, the architecture fails in one of two directions: flat (B3.03) when B1.02 is absent, or intrinsically typed (B3.18) when B1.17 is absent. Only with both commitments active does the three-level structure serve the governance purpose for which it was introduced.

The next composition pair note, B4.05, addresses B1.02 + B1.20 — how three-level structure and recursive governance combine so that the Paper 1 commitments that apply at every level are inherited consistently across the structure the present note explains how to populate.

Source paper

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

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